

Rodrigo Sierra

Applied AI Portfolio

Process optimization • AI automation • Business systems

Houston City College | A.A.S. Artificial Intelligence

AI is not only about code. It is about solving real problems with models, tools, workflows, and human-centered decisions.

Portfolio Theme

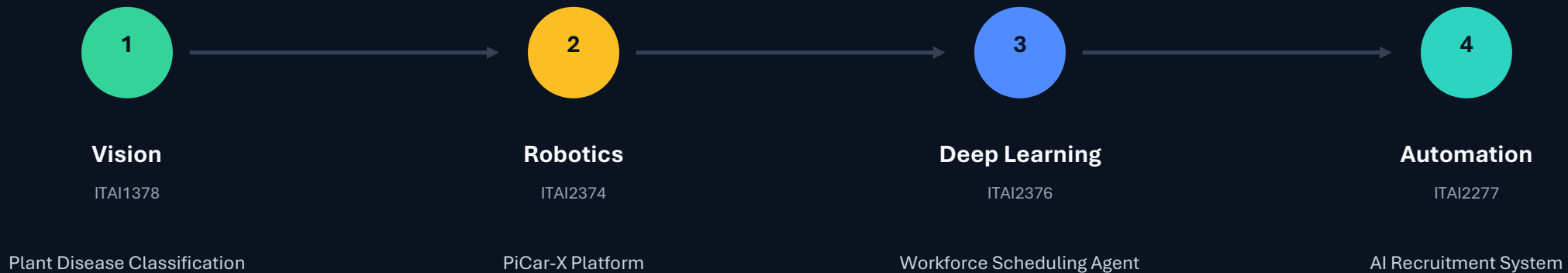
Applied AI for Process Optimization

GitHub Portfolio

github.com/RodMLGH/Rodrigo-Sierra-AI-Portfolio

Learning Journey Across Applied AI

Four projects show a progression from model use to integrated AI systems.



AI maturity shown: image models → robotic platforms → forecasting + optimization → end-to-end business automation.

Featured Projects

Selected work demonstrates model development, robotics, agents, optimization, and business workflow integration.

1

Plant Disease Classification

Computer vision model for leaf-based disease detection.

PyTorch

ResNet50

CV

2

PiCar-X Robotics Project

Hands-on robotics with sensors, actuators, and platform integration.

Robotics

Sensors

Testing

3

Workforce Scheduling Agent

Multi-agent system for demand forecasting, scheduling, validation, and explanation.

CrewAI

LSTM

OR-Tools

4

AI Recruitment System

n8n workflow integrating email, database, calendar, and LLM evaluation.

n8n

OpenAI

Airtable

Technical Arsenal

A practical skill set for connecting AI capabilities to operational needs.

Build Python • PyTorch • TensorFlow/Keras • pandas

Optimize LSTM forecasting • OR-Tools scheduling • validation

Integrate n8n • Airtable • Gmail • Calendar • APIs

Explain OpenAI • Gemini • business-friendly outputs

Core competency: translate real-world processes into AI-enabled systems that improve work, decisions, and value.

Direction: AI for Process Optimization

My focus is applying AI where operations, people, data, and systems meet.

Where I add value

- Business process analysis.
- Workflow automation.
- Decision-support systems.

Target domains

- Operations & logistics.
- HR and recruitment.
- Manufacturing & services.

Next steps

- OPT-aligned AI roles.
- Stronger project documentation.
- Houston-area industry applications.

Portfolio link

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Thank you